Note 12 Regression with Autocorrelated Errors

1 Review

1.1 Simple Linear Regression

In note 4, we have briefly introduced the classical regression model with uncorrelated errors, the simple linear regression model.

$$Y_t = \beta_0 + \beta_1 X_t + \epsilon_t.$$

- Y_t is the response variable and X_t is the predictor variable.
- β_0 is the intercept and β_1 is the slope. They can be estimated by ordinary least square method. That is, $\hat{\beta}_0$ and $\hat{\beta}_1$ are obtained by minimizing

$$SSE = \sum_{t=1}^{N} (Y_t - \beta_0 - \beta_1 X_t)^2$$

- $\{\epsilon_t\}$ are **independent** identically distributed random errors.

1.2 Multiple Linear Regression

Multiple linear regression model describe the relation between a single response variable Y_t and multiple predictor variables.

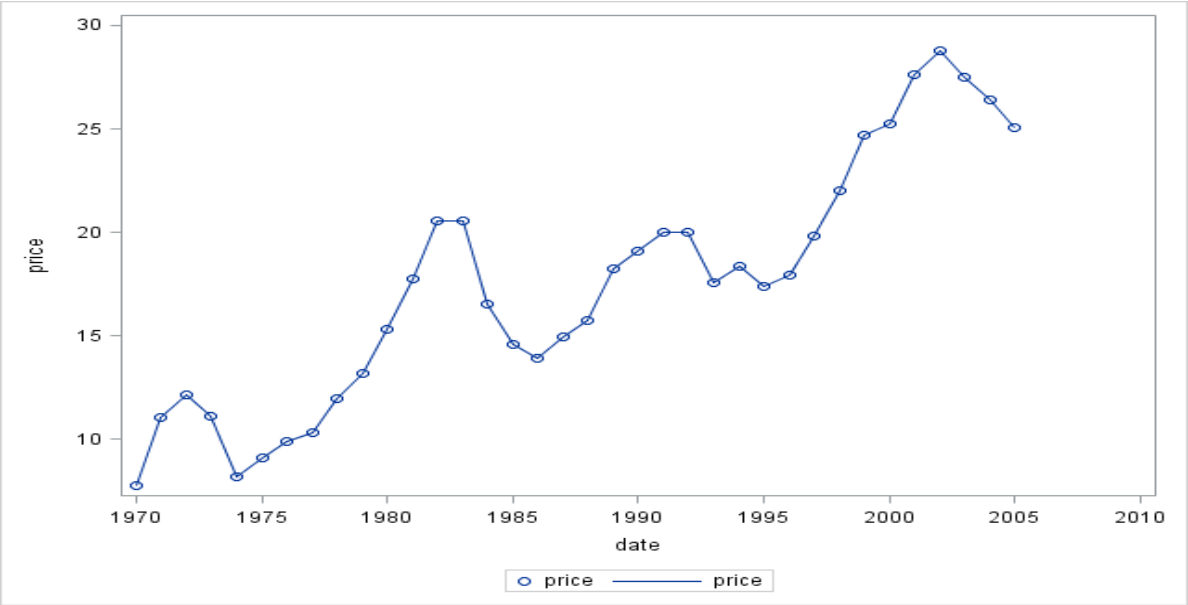
$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \epsilon_t.$$

Similarly, the parameters are obtained by minimizing

$$SSE = \sum_{t=1}^{N} (Y_t - \beta_0 - \beta_1 X_{1t} - \cdots - \beta_k X_{kt})^2.$$

$\{\epsilon_t\}$ are **independent** identically distributed random errors.

Example 1.1. The price of a precious metal (dollars per gram) from 1970 to 2005 is plotted below.



Fit a regression model to predict the price in 2006.

1.3 Durbin-Watson Statistics

The Durbin-Watson statistics is

$$D = \frac{\sum_{t=2}^{N}(r_t - r_{t-1})^2}{\sum_{t=1}^{N} r_t^2},$$

where r_t are the residuals, $r_t = Y_t - \hat{\beta}_0 - \hat{\beta}_1 X_{1t} - \cdots - \hat{\beta}_k X_{kt}$.

- $0 \leq D \leq 4$
- If the residuals are white noise, D is close to 2.

2 Regression with Autocorrelated Errors

When we do regressions using time series variables, it is common for the errors (residuals) to have a time series structure. This violates the assumption of independent errors. The consequence is that the estimates of coefficients and their standard errors will be wrong if the time series structure of the errors is ignored.

Regression with autocorrelated error models have the following form:

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \epsilon_t,$$

where the $\{\epsilon_t\}$ is modeled as a ARMA process.

2.1 General Steps

1. Initially, assume the random errors $\{\epsilon_1, \epsilon_2, \cdots, \epsilon_N\}$ are independent, estimate the parameters $\beta_0, \beta_1, \beta_k$ be ordinary least square estimation.

2. After fitting the regression model, calculate the residuals

   $$r_t = Y_t - \hat{Y}_t = Y_t - (\hat{\beta}_0 + \hat{\beta}_1 X_{1t} + \cdots + \hat{\beta}_k X_{kt})$$

   Since the random error ϵ_t are not observed, r_t are estimates of ϵ_t.

   - If residuals seem independent and all diagnostic tests are passed, we can stop here. It is not necessary to have a regression model with autocorrelated errors.
   - If the residuals are not independent, continue with the following steps.

3. (optional) If trend exists in $\{r_t\}$, difference both the response variable Y_t and the predictor variables $X_{1t}, \cdots, X_{kt}$ and fit a regression model using the differenced variables. That is,

   $$\nabla Y_t = \beta_0 + \beta_1 \nabla X_{1t} + \beta_2 \nabla X_{2t} + \cdots + \beta_k \nabla X_{kt} + \epsilon_t$$

   Then estimate the parameters and calculate the residuals

   $$r_t = \nabla Y_t - (\hat{\beta}_0 + \hat{\beta}_1 \nabla X_{1t} + \cdots + \hat{\beta}_k \nabla X_{kt})$$

4. (optional) If seasonal effect exists in $\{r_t\}$, difference both response variable Y_t and the predictor variables $X_{1t}, \cdots, X_{kt}$ at lag of the period T and fit a regression model using the differenced variables. That is,

   $$\nabla^T \nabla Y_t = \beta_0 + \beta_1 \nabla^T \nabla X_{1t} + \beta_2 \nabla^T \nabla X_{2t} + \cdots + \beta_k \nabla^T \nabla X_{kt} + \epsilon_t$$

   Then estimate the parameters and calculate the residuals

   $$r_t = \nabla^T \nabla Y_t - (\hat{\beta}_0 + \hat{\beta}_1 \nabla^T \nabla X_{1t} + \cdots + \hat{\beta}_k \nabla^T \nabla X_{kt})$$

5. Fit an ARMA model to the residuals $\{r_t\}$. That is,

$$\phi(B)r_t = \theta(B)Z_t \qquad \text{or} \qquad r_t = \frac{\theta(B)}{\phi(B)}Z_t$$

where $\{Z_t\}$ is Gaussian white noise with mean zero and variance σ_Z^2.

6. The regression model becomes

$$\nabla^T\nabla Y_t = \beta_0 + \beta_1\nabla^T\nabla X_{1t} + \beta_2\nabla^T\nabla X_{2t} + \cdots + \beta_k\nabla^T\nabla X_{kt} + \frac{\theta(B)}{\phi(B)}Z_t \qquad (1)$$

Use maximum likelihood method to estimate the parameters in this model.

Note: Most of the time, step 3 and 4 are omitted. Then (1) is reduced to

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \frac{\theta(B)}{\phi(B)}Z_t$$

7. Diagnostics.

8. Forecasting.

Example 2.1. Use example 1.1, fit a regression model with autocorrelated errors.

3 Diagnostics

4 Forecast